

X A - B E N - N A - P O
BIG VALLEY BAND OF POMO INDIANS



Clear Lake Cyanotoxin Monitoring Program

The Clear Lake Cyanotoxin Monitoring Program <https://bit.ly/ClearLakeCyanoProgram> which was developed by Big Valley and Elem Indian Colony in 2014 continued with its regular monitoring of the Clear Lake shoreline in 2024. At this time, the program is funded by Tribal and federal sources. Monthly monitoring of fewer shoreline sites occurred during January – April. Beginning in May because of bloom activity, the program started monitoring biweekly sampling for all regularly monitored sites. At all locations, we collect water chemistry data, phycocyanin and chlorophyll-a, and take samples for microscopy and cell identification. All data is submitted to the Water Quality Exchange. This year the Tribe is submitted impairment status in ATTAINS, along with a number of Tribes in the US, and the States. We did not have any detects of toxins other than Microcystin although Saxitoxin and Anatoxin-a gene copies were present (identified by Quantitative polymerase chain reaction - qPCR). The highest toxin level highest result value on Clear Lake was on June 25th with a **16,025 µg/L** at Lily Cove, a residential location and private Homeowners Association shoreline in the Lower Arm of Clear Lake. This result value is 1,000 times higher than the Danger trigger level for Microcystins. The cyanobacteria blooms varied throughout the year, with Microcystis cyanobacteria as the dominant genus in 27% of our samples, nitrogen-fixing cyanobacteria such as Dolichospermum, Aphanizomenon, Gloeotrichia, Anabaenopsis and Cuspidothrix spp in 30.6% of the samples, and other non-nitrogen fixing cyanobacteria in 41.5% of our samples. The California Source Water Ambient Monitoring Program maintains a chart of Cyanobacteria and their most common toxins <https://bit.ly/CyanobacteriaToxinsChart>

In 2021 the team of Tracking California, OEHHA, Waterboards Drinking Water Division and Big Valley started the Cal-WATCH tap water monitoring program which became an extremely useful tool that found that more than 400 at risk private/self-supplied drinking water systems were pulling water directly from Clear Lake. Because of the unprecedented Microcystin and Anatoxin-a toxin levels on Clear Lake at that time, Lake County Public Health issued a Health Advisory on September 16, 2021 of Do Not Drink the tap water for these individual homes

(specifically in the Oaks and Lower arms) that draw their drinking water from Clear Lake. Alternate sources of tap water were set up for these homes. In 2024, we assessed 2 homes with lake intakes for cyanotoxins and cyanobacteria and found cyanobacteria in the treated tap water of both, and toxins in 1 of them. US EPA's 2015 Drinking Water Health Advisory for Microcystins threshold sets 0.3 µg/L for Public Water Systems. <https://bit.ly/EPAMicrocystinHealthAdvisory>

Both of the participants we tested this year were counseled to get their drinking water systems serviced, and to not drink the water until new testing shows no detect of cyanobacteria. Here's some of the outreach provided to the participants with cyanobacteria or toxins in their tap water. www.cal-watch.org

Cal-WATCH Program

Steps you should take based on cyanotoxin testing results.

Cyanotoxins

We found cyanobacteria cells, cyanotoxin, or both in your water sample.

We detected cyanobacteria cells and/or one or more cyanotoxins above levels of concern (the US EPA drinking water health advisory for [microcystin/cylindrospermopsin](#) or the [World Health Organization provisional health-based reference value for anatoxin-a](#)). This means there is the potential for human or animal illness from consuming your water.

We recommend that you stop drinking or cooking with tap water until the problem is resolved.

- Use **bottled water or an alternative water source** for drinking and cooking
- **Do not drink or cook with boiled tap water** (Boiling water concentrates the cyanotoxins)
- Bathing or showering in tap water is safe for children over age 6 and adults
- Washing clothes and dishes in tap water is safe
- **Do not start drinking and cooking with your tap water again until you have retested and have been told your drinking water is safe**

We suggest you **keep the attached water testing results for your files**, to show anyone who is working on your water treatment system.

More information on **organizations and companies who can help you** can be found on page 4 of the packet.

Maximum Microcystin Value on Clear Lake since 2014

Ten out of the 11 years of Clear Lake monitoring had danger levels of microcystin in at least one location, and six of those ten were in the Lower Arm of the lake. The Danger trigger level is defined in the Six Cyanotoxins Study conducted by CalEPA's Office of Environmental Health Hazard Assessment as 20 ug/L. All maximum Danger levels on Clear Lake have been at least ten times higher, with most being 100-1,000 times higher than the trigger level.

Year	Maximum Microcystin Value (µg/L)	Location	Arm of Lake
2024	16,025	LC01	L
2023	16,821	JB	L
2022	790	SBMMEL01	O
2021	160,378	RED01	L
2020	1,146	LUC01	U
2019	150	LC01	L
2018	4,800	SBMMEL01	O
2017	5,554	AP01	L
2016	0.67	SBMMEL01	O
2015	10,162	AP01	L
2014	16,920	CLOAKS01	O

Summertime Monitoring 2024 results

For the purposes of looking at Clear Lake for its recreational and cultural uses, the review of the summertime toxins starts at the June 10, 2024, sampling event and ends with the October 2, 2024, sampling event. Fifteen of the regularly sampled twenty-three sites had Caution, Warning or Danger levels at least 50% of the time. Caution signage recommends to not eat shellfish from those sites, and to prevent children and animals from being in discolored waters/bloom areas. Seven of those fifteen sites were in the Lower Arm of Clear Lake.

SITE ID	ARM	6/10	6/25	7/10	8/6	8/20	9/4	9/18	10/2	SAMPLING EVENTS	# CAUTION	# WARNING	# DANGER	SUMMER % TIMES AT C/W/D (6/10-10/2)
AP01	L	DANGER	DANGER	DANGER	DANGER	WARNING	WARNING	CAUTION	CAUTION	8	2	2	4	100%
BP	L	NONE	CAUTION	CAUTION	CAUTION	CAUTION	CAUTION	WARNING	NONE	8	5	1	0	75%
BVCL6	U	NONE	NONE	NONE	CAUTION	NONE	CAUTION	NONE	NONE	8	2	0	0	25%
CL-1	U	NONE	N/A	NONE	N/A	N/A	NONE	N/A	N/A	3	0	0	0	0%
CL-3	L	CAUTION	N/A	CAUTION	N/A	N/A	WARNING	N/A	N/A	3	2	1	0	100%
CL-4	O	CAUTION	N/A	CAUTION	N/A	N/A	CAUTION	N/A	N/A	3	3	0	0	100%
CLOAKS01	O	NONE	DANGER	DANGER	DANGER	WARNING	CAUTION	NONE	CAUTION	8	2	1	3	75%
CLV7	U	NONE	WARNING	WARNING	WARNING	NONE	NONE	NONE	NONE	8	0	3	0	38%
CP	U	NONE	NONE	NONE	CAUTION	NONE	CAUTION	NONE	NONE	8	2	0	0	25%
FC3	U	NONE	WARNING	WARNING	N/A	N/A	N/A	N/A	N/A	3	0	2	0	67%
ELEM01	O	CAUTION	N/A	DANGER	WARNING	WARNING	WARNING	NONE	NONE	7	1	3	1	71%
GH	O	CAUTION	DANGER	WARNING	CAUTION	CAUTION	CAUTION	NONE	CAUTION	8	5	1	1	88%
HB	U	N/A	CAUTION	CAUTION	CAUTION	NONE	NONE	NONE	CAUTION	7	4	0	0	57%
JB	L	DANGER	CAUTION	WARNING	WARNING	WARNING	WARNING	DANGER	CAUTION	8	2	4	2	100%
KEYS03	O	NONE	DANGER	DANGER	DANGER	DANGER	DANGER	WARNING	CAUTION	8	1	1	5	88%
KP01	U	NONE	CAUTION	DANGER	NONE	NONE	NONE	NONE	CAUTION	8	2	0	1	38%
LC01	L	NONE	DANGER	DANGER	DANGER	CAUTION	DANGER	CAUTION	CAUTION	8	3	0	4	88%
LPTNT	U	NONE	NONE	DANGER	NONE	NONE	CAUTION	CAUTION	NONE	8	2	0	1	38%
LUC01	U	NONE	NONE	NONE	NONE	NONE	CAUTION	NONE	NONE	8	1	0	0	13%
RED01	L	DANGER	DANGER	DANGER	WARNING	WARNING	DANGER	CAUTION	CAUTION	8	2	2	4	100%
RODS	U	NONE	NONE	NONE	N/A	NONE	CAUTION	NONE	NONE	7	1	0	0	14%
SBMMEL01	O	NONE	DANGER	DANGER	WARNING	WARNING	DANGER	NONE	NONE	8	0	2	3	63%
SHADY01	L	CAUTION	WARNING	CAUTION	CAUTION	CAUTION	WARNING	CAUTION	CAUTION	8	6	2	0	100%

The breakdown of Microcystin toxin levels by arm in summer 2024 is as follows:

LOWER ARM			
SUMMER CAUTION	SUMMER WARNING	SUMMER DANGER	TOTAL SUMMER EVENTS
22	12	14	51
43%	24%	27%	94%
OAKS ARM			
SUMMER CAUTION	SUMMER WARNING	SUMMER DANGER	TOTAL SUMMER EVENTS
12	8	13	42
29%	19%	31%	79%
UPPER ARM			
SUMMER CAUTION	SUMMER WARNING	SUMMER DANGER	TOTAL SUMMER EVENTS
14	5	2	68
21%	7%	3%	31%

The 7 Lower Arm sites had more Caution toxin levels (although it had the highest toxin level on the lake in 2024). 94% of the sampling events were above the Caution trigger level.

The 6 Oaks Arm sites had more Danger toxin levels than Caution and Warning. 79% of the sampling events were above the Caution trigger level.

The Upper Arm had more Caution toxin levels than Warning and Danger. 31% of the sampling events were above the Caution Trigger level.

CAUTION - PRECAUCIÓN

Harmful algae may be present in this water.
Puede haber algas dañinas en estas aguas.

 Stay away from algae and scum in the water. Aléjese de las algas o espuma lamosa en el agua.	 Do not let pets and other animals go into or drink the water, or eat scum on the shore. No deje que sus mascotas o animales se metan o beban el agua, o se acerquen a la espuma lamosa.
 Keep children away from algae in the water or on the shore. Mantenga a los niños alejados de algas en el agua u orilla del agua.	 Do not drink this water or use it for cooking. No beba de esta agua o use para cocinar.
 For fish caught here, throw away guts and clean fillets with tap water or bottled water before cooking. Al pescado que pesque aquí, quítele los intestinos y tírelos a la basura. Limpie el filete con agua de la llave o embotellada antes de cocinarlo.	 Do not eat shellfish from this water. No coma mariscos de estas aguas.

Call your doctor or veterinarian if you or your pet get sick after going into the water.
Lláme a su médico o veterinario si usted o su mascota se enferman después de meterse al agua.
For more information go to (Para información): <https://mywaterquality.ca.gov/habs/>
For local information, contact (Para información local comuníquese con):

Caution levels of Microcystin are 0.8 ug/L – 5.99 µg/L.

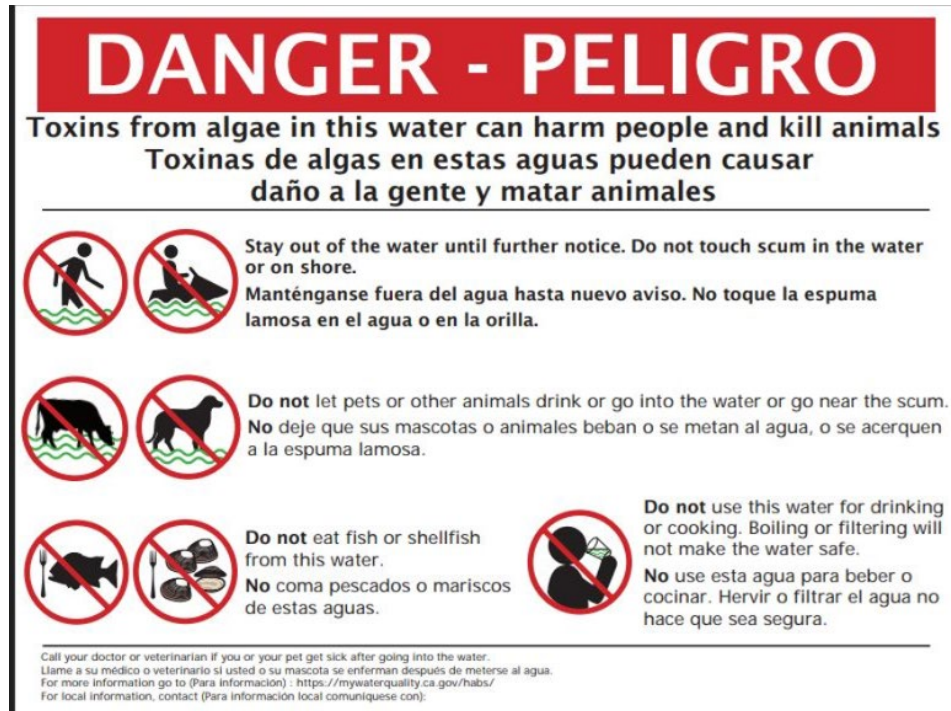
WARNING - ADVERTENCIA

Toxins from algae in this water can harm people and kill animals
Toxinas de algas en estas aguas pueden causar daño a la gente y matar animales

 No swimming. Prohibido nadar.	 Do not let pets or other animals go into or drink the water, or go near the scum. No deje que sus mascotas o animales se metan o beban el agua, o se acerquen a la espuma lamosa.
 Stay away from scum, and cloudy or discolored water. Manténgase alejado de la espuma lamosa, y agua turbia o descolorida.	 Do not eat shellfish from this water. No coma mariscos de estas aguas.
 Do not use this water for drinking or cooking. Boiling or filtering will not make the water safe. No use esta agua para beber o cocinar. Hervir o filtrar el agua no hace que sea segura.	 For fish caught here, throw away guts and clean fillets with tap water or bottled water before cooking. Al pescado que pesque aquí, quítele los intestinos y tírelos a la basura. Limpie el filete con agua de la llave o embotellada antes de cocinarlo.

Call your doctor or veterinarian if you or your pet get sick after going into the water.
Lláme a su médico o veterinario si usted o su mascota se enferman después de meterse al agua.
For more information go to (Para información): <https://mywaterquality.ca.gov/habs/>
For local information, contact (Para información local comuníquese con):

Warning levels of Microcystin are 6 ug/L – 19.99 µg/L.



Danger levels of Microcystin is 20 µg/L and above.

Clear Lake Cyanobacteria and Microscopy

Big Valley EPA staff look at all water samples to determine dominant and secondary cyanobacteria genera. Because each cyanobacteria genus can produce specific toxins, the program uses microscopy to assess what toxins to analyze for. The trends we saw in 2024 show that by mid-April, cyanobacteria made up more than 75% of the samples during a sampling event. Nitrogen fixing cyanobacteria dominated the samples early in the season. Microcystis sp. were dominant in the lake over all other genera only in late May/early June. Nitrogen fixing cyanobacteria dropped while Microcystis and other non-nitrogen fixing cyanobacteria dominated throughout the summer.

